CDE Enterprise Architecture

End-to-End Systems, AI Intelligence & Observability

Complete diagrams • Aspect-ratio preserved • Submission edition

Reading order

01 Technology Stack → 02 Taylor IoT Integration → 03 SICC & Vizzio

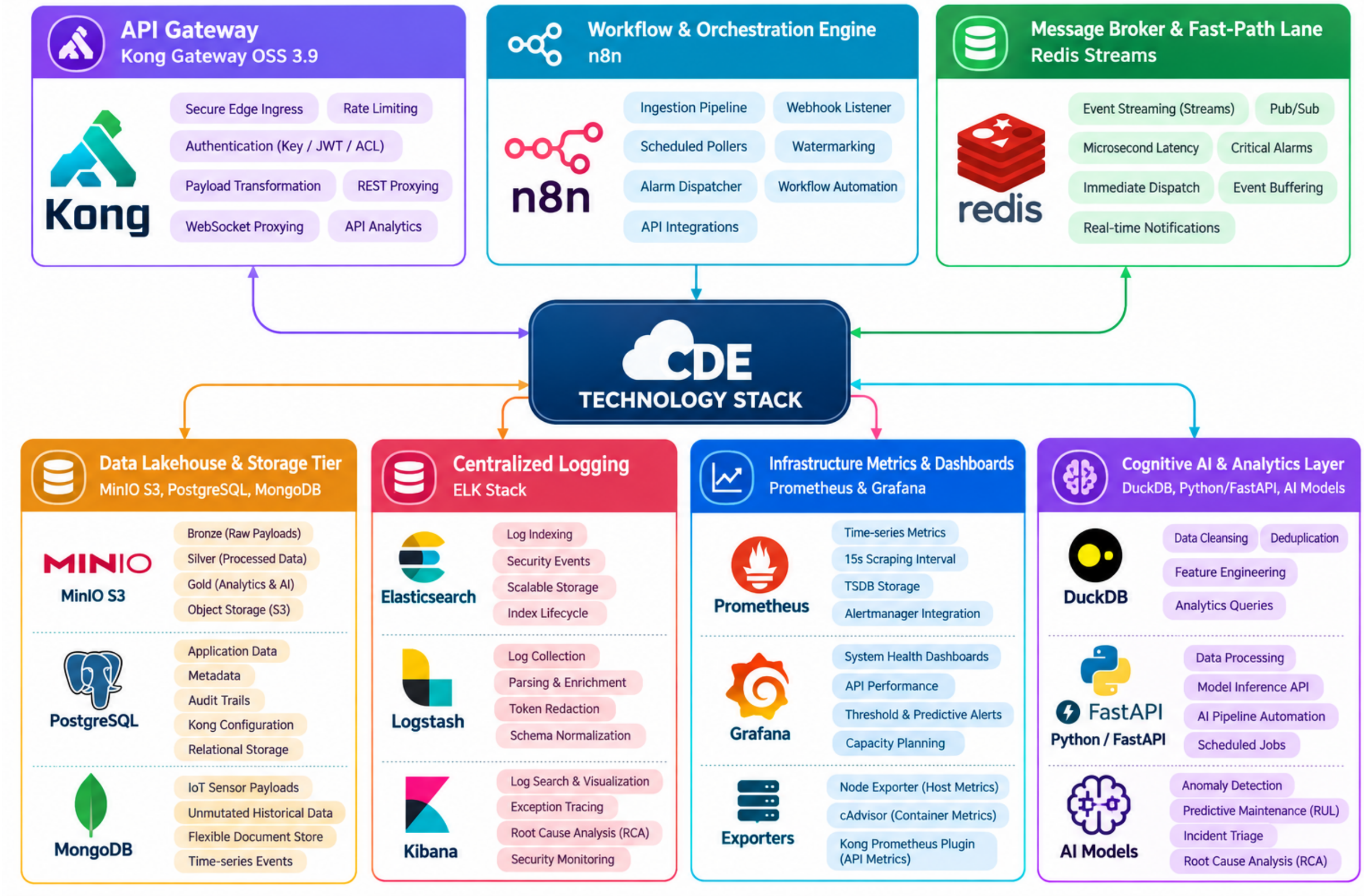
04 Enterprise CDE → 05 Cognitive AI → 06 Observability

07 Closed-loop AIOps Advisor

Design note: Diagram pages use individually adapted dimensions so no original image is cropped.

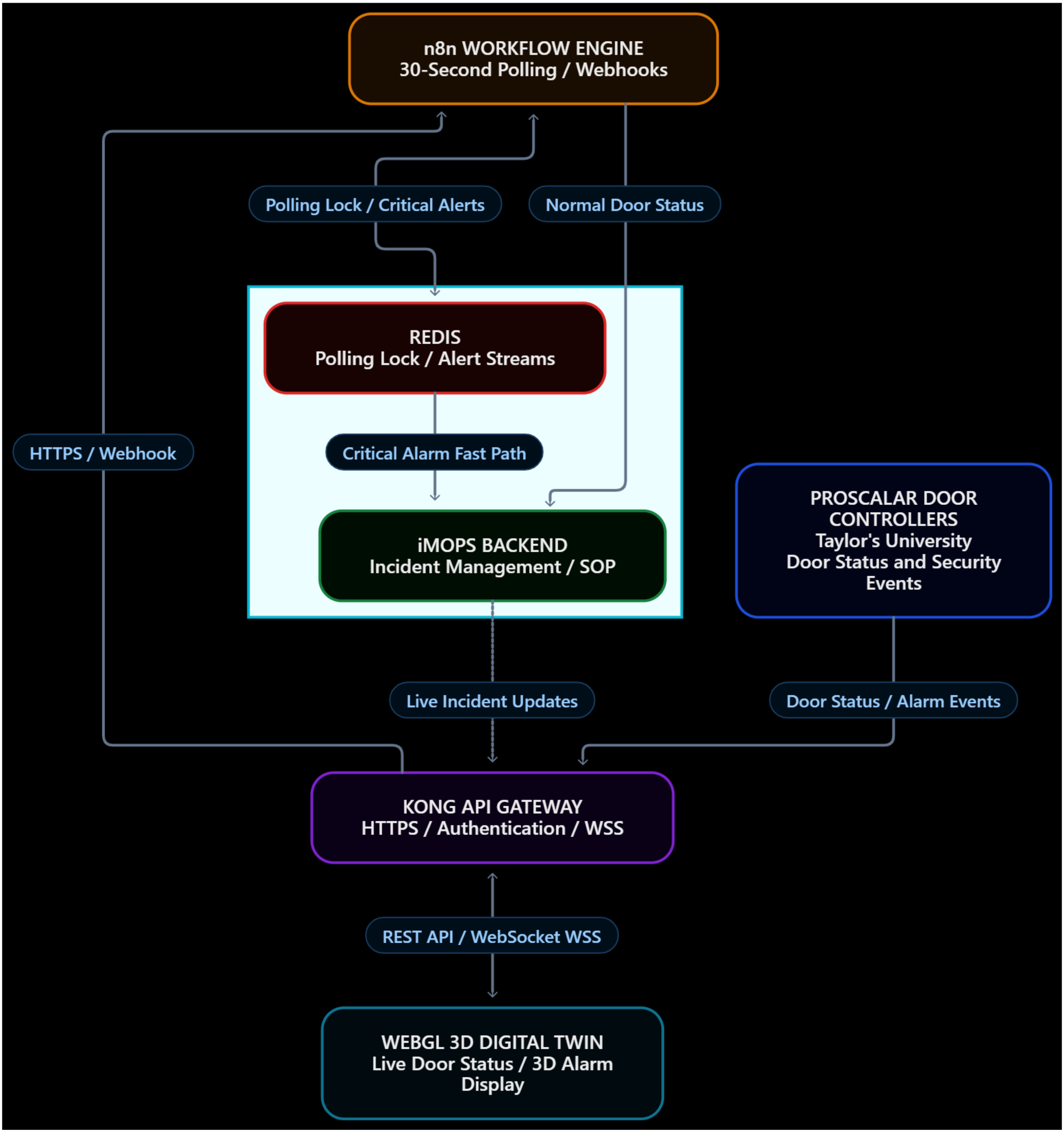
01 Technology Stack

Core components, responsibilities and platform services.



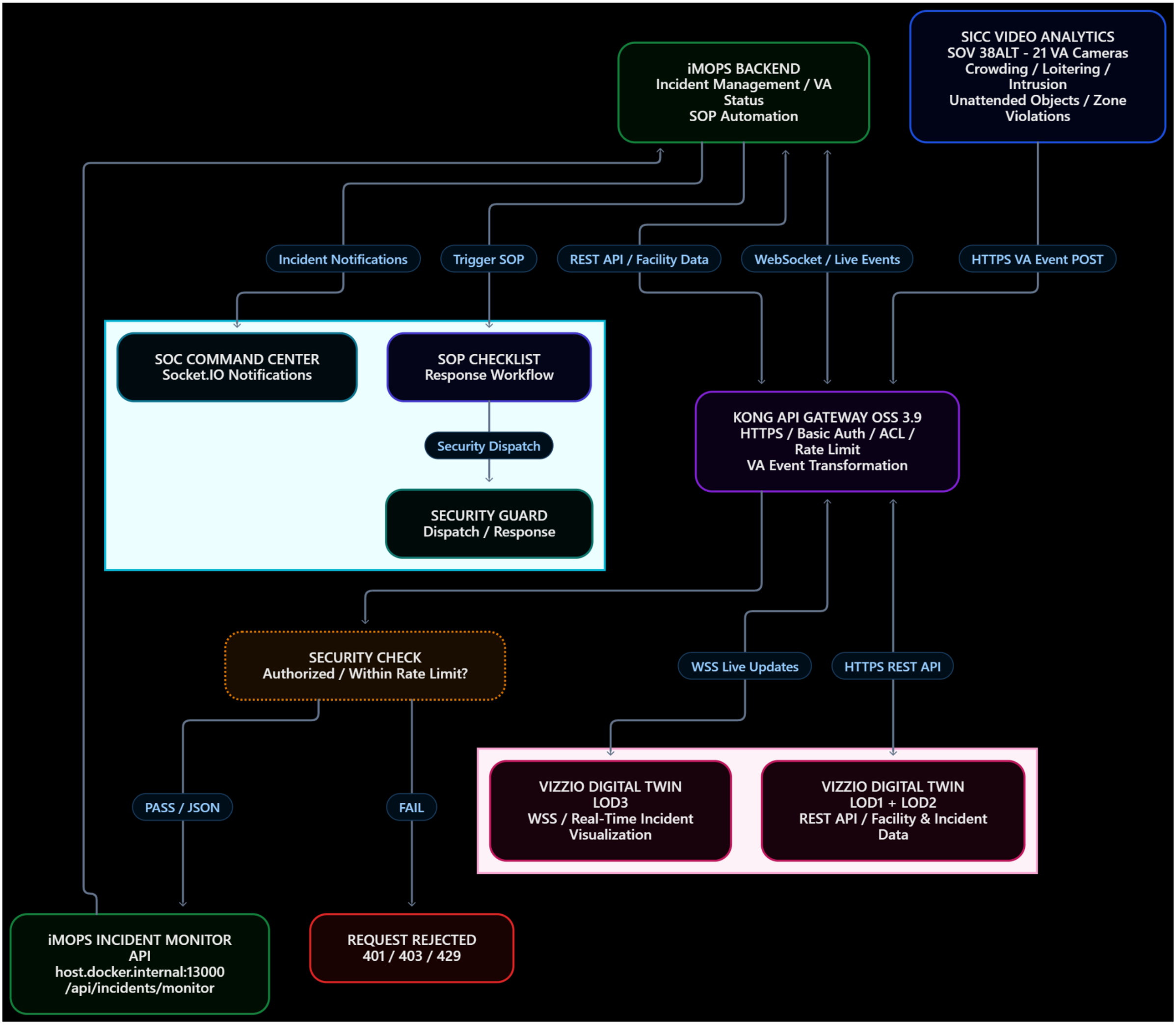
02 Taylor IoT Integration

Door controllers, 30-second polling, Redis alerts and WebGL updates.



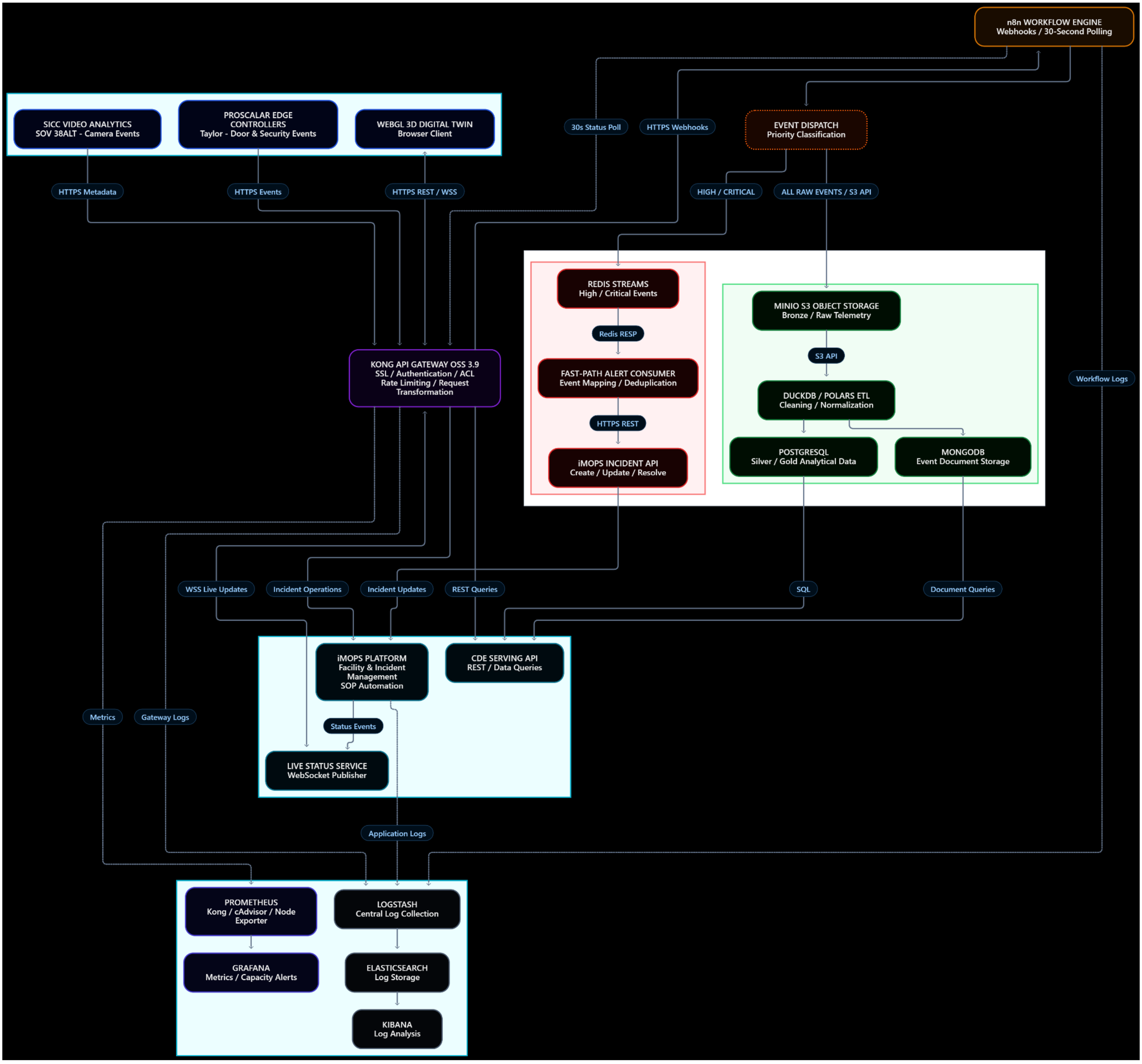
03 SICC Video Analytics & Vizzio

21 camera feeds, Kong ingress, iMOPS, REST LOD1/2 and WSS LOD3.



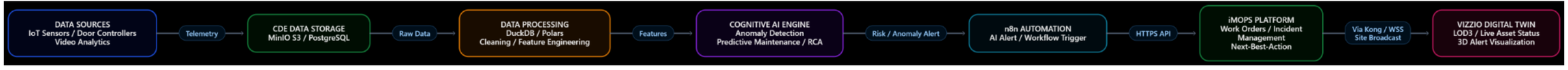
04 Enterprise CDE End-to-End

Dual-path alert and lakehouse processing, serving APIs and observability.



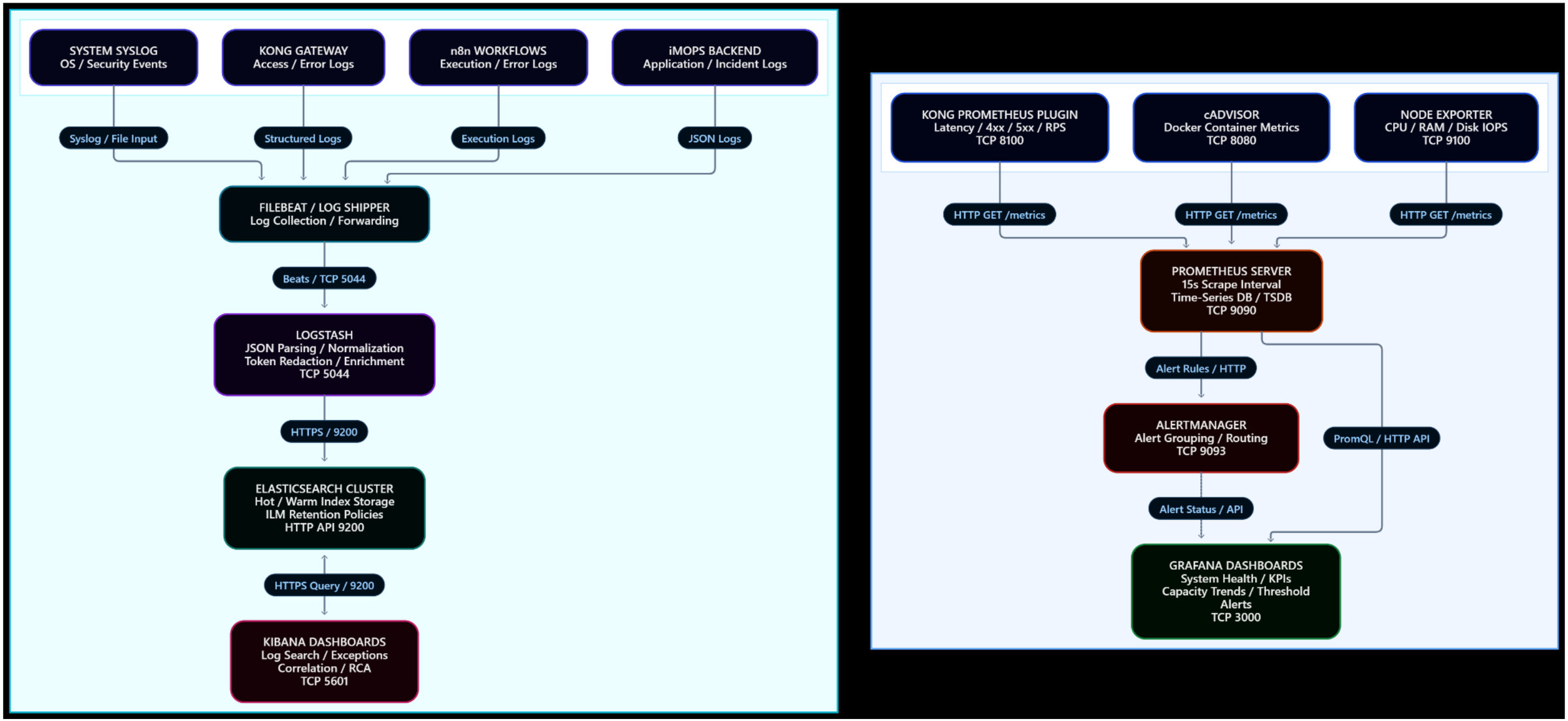
05 Data Processing & Cognitive AI

From raw data ingestion to AI insights and Digital Twin actions.



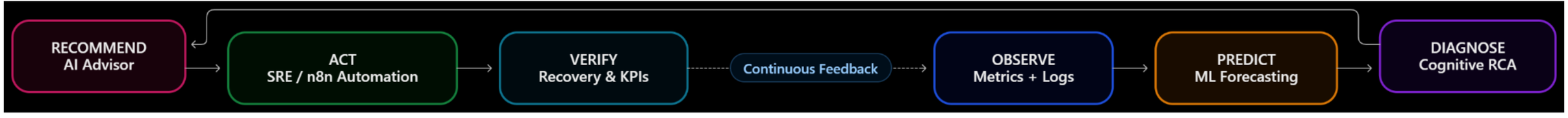
06 Centralized Observability

Metrics and logs, Prometheus/Grafana and ELK.



07 Closed-Loop Cognitive Advisor

Observe, predict, diagnose, recommend, act and verify.



Architecture Sequence & Integration Notes

- **1. Ingestion and routing**

  External sensors and VA sources submit events through Kong; orchestration accepts and routes them.

- **2. Independent data paths**

  High-priority alarms enter Redis Streams; all raw payloads enter durable lakehouse storage.

- **3. Incident response**

  A consumer creates/updates iMOPS incidents and drives SOC notifications and approved SOP workflows.

- **4. Digital Twin subscriptions**

  LOD1/LOD2 queries use REST; LOD3 receives site-scoped WSS broadcasts via Kong after subscription.

- **5. Data intelligence**

  ETL prepares historical data; predictive analytics generates risk findings and recommended actions.

- **6. Monitoring and feedback**

  Prometheus/Grafana tracks metrics and ELK indexes logs; results inform RCA and capacity forecasts.

- **7. Governed closed loop**

  AI recommendations trigger alerts/tickets; only approved runbooks execute, then recovery is verified.

Logical design only: performance targets, causal findings, and operational workflows require implementation validation.